SAME SHADOWS, DIFFERENT FORESTS: LOCAL PREDECESSOR STRUCTURE AND GLOBAL DYNAMICS OF COLLATZ-TYPE MAPS

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Abstract. What determines the global dynamics of a Collatz-type map when its local predecessor structure is indistinguishable from that of another map? We study this question for the maps $x \mapsto (qx + c)/2$ and give three complementary answers. First, every local statistic of the predecessor trees of $3x+1$ (branching classes, Leftward and Upward children, cotree densities, densities of finite parity patterns) coincides with that of $3x - 1$, which is $3x + 1$ on the negative integers and has three cycles. Local branching statistics therefore determine neither the cycle structure nor basin membership. Second, cycles are characterised by an exact global condition: a parity word B of length l with k odd steps yields an integer cycle if and only if $D = 2^l - q^k$ divides $c\, z_q(B)$. Taking the reverse perspective of rational cycles, we ask for which maps a given word yields a cycle. One pass over all $2.2 \cdot 10^{12}$ words of length at most 40 then gives the complete list of 1097 primitive cycles of all maps $3x + c$ with $|c| \leq 1000$. For $0 < c < 1000$ this list agrees exactly with the conjecturally complete tables of Belaga, Mignotte and Morley (OEIS A226607) for all cycles with at most 40 elements. An equidistribution heuristic predicts 1149 cycles, and the observed counts scatter less than a Poisson model allows. Third, every basin of attraction of a map $3x + c$ with odd $c \geq -1$ has a logarithmic density. This follows from the stabilisation of first passages due to Tao and to Gonçalves, Greenfeld and Madrid, because a basin is not changed by first passage. A divisibility filtration explains certain exact rational basin densities. The remaining densities, measured up to 10^{10} for 18 maps, are stable to three digits but unexplained: in $3x + 5$, two cycles of the same type attract about 50 % and 9 % of all integers.

Date: October 2026.

2020 *Mathematics Subject Classification.* Primary 11B83; Secondary 11Y55, 37P99.

Key words and phrases. Collatz problem, $qx+1$ problem, cycles, parity vectors, rotating binaries, Diophantine approximation, basins of attraction, Cayley trees.

1. Introduction

Crandall [1] generalized the $3x+1$ sequence by introducing the function $f : \mathbb{N} \to \mathbb{N}$,

$$f(x) = \begin{cases} qx+1 & 2 \nmid x, \\ x/2 & \text{otherwise}, \end{cases} \tag{1}$$

for odd $q \geq 3$. More generally, for odd $q \geq 1$ and odd c we consider $x \mapsto qx+c$. Since every odd step is followed by at least one halving, we work with the accelerated map

$$T_{q,c}(x) = \begin{cases} (qx+c)/2 & 2 \nmid x, \\ x/2 & \text{otherwise}, \end{cases} \tag{2}$$

which has the same cycles as (1) (for $c = 1$), up to the omitted even members. We call a "$qx+c$ cycle" a sequence of distinct positive integers $(x_1, x_2, \ldots, x_n)$ with $T_{q,c}(x_i) = x_{i+1}$ and $T_{q,c}(x_n) = x_1$.

Leslie Davison observed that *circuits*, i.e. cycles consisting of one run of odd steps followed by one run of halvings, correspond one-to-one to the solutions (h, k, l) of the Diophantine equation

$$(2^{k+l} - 3^k)h = 2^l - 1, \tag{3}$$

see [2, p. 67]. Steiner [3] proved that $(h, k, l) = (1, 1, 1)$ is the only solution, so the trivial cycle is the only circuit of $3x+1$. Simons and de Weger [4] and Hercher [5] extended this to m-cycles (cycles with m runs of odd steps) for $m \leq 68$ and $m \leq 91$, respectively. The general cycle problem remains open; convergence has been verified for all $n < 2^{71}$ [6].

For $q > 3$ cycles are rare. Table 1 lists the non-trivial cycles of (1) for odd $q \leq 1000$.

q	**odd members of the cycle**
5	$13, 33, 83$
5	$17, 27, 43$
181	$27, 611$
181	$35, 99$

Table 1. Non-trivial $qx+1$ cycles for odd $q \leq 1000$. Their role for Crandall's conjecture was studied by Franco and Pomerance [7]; Koch et al. [8] confirmed them in a data-science search. A search over all odd $q \leq 1000$ and all odd starting values below 3000 finds exactly these, besides the cycles through 1 that exist for $q = 5$ and for $q = 2^m - 1$.

Let $(a_1, a_2, \ldots, a_n)$ be the odd members of a cycle of (1) in cyclic order. Then the product

$$2^\alpha = \prod_{i=1}^{n} \left(q + \frac{1}{a_i} \right) \tag{4}$$

is a power of two, see for example [2, p. 327]. Here α is the number of divisions by 2 in one period, and n is the number of odd members.

Related work. Cycles of the maps $3x + d$ have been tabulated before. Lagarias [9] introduced primitive cycles, whose members are coprime to d, and conjectured that every map $3x+d$ has at least one and only finitely many of them. Belaga and Mignotte [10, 11] listed the primitive cycles for all $d < 20000$, completely as they argue, and Morley [12] confirmed these lists by following every trajectory with starting value below 10^8. Belaga and Mignotte also embedded the $3x+1$ conjecture in the $3x+d$ context [13] and described the cycles of all maps $3x+d$ at once, through a duality between the shift d and the members of a cycle [14]. Cox [15] modelled the number of cycles per d by a Poisson distribution. These lists are conjectural: a cycle whose basin contains no small starting value would be missed. The census of Section 4 is complete by construction for bounded word length, and it confirms the published lists in that range.

Densities enter through predecessor sets. Wirsching [16] studied the density of the set of integers that iterate to a fixed integer under $3x+1$ with 3-adic methods. Tao [17] proved that almost all orbits attain almost bounded values, in the sense of logarithmic density, and Gonçalves, Greenfeld and Madrid [18] extended this to a class of maps that contains $3x + c$ for odd $c \geq -1$. They also observed numerically that the three cycles of $3x - 1$ attract about a third of all integers each, and asked whether such densities exist [18, Remark 1.7]; their question concerns natural densities. Mazur [19] recently upgraded Tao's theorem to natural density, with an explicit logarithmic time bound, by comparing the first-passage laws of natural and logarithmic counting.

On the local side, Kleinnijenhuis, Kleinnijenhuis and Aydogan [20] develop a graph-theoretic approach to the $an+b$ conjectures. They represent these maps by Cayley trees with Leftward and Upward edges and three-colour pictures, with a detailed congruence-class bookkeeping that includes cotree densities. Section 2 adopts this tree perspective.

Two views of the same maps. These maps admit two descriptions. The *local* view starts from the trivial cycle and grows the tree of all predecessors, organised by congruence classes. It describes how numbers branch, and it is the same for every map in the family: by $x \mapsto -x$, the $3x - 1$ map on the positive integers is the $3x + 1$ map on the negative integers, which has the three cycles through -1, -5 and -17. Both maps produce the same branching statistics (Proposition 2). In the language of Plato's cave,

they cast identical shadows. Density arguments built on this view reach "almost all" statements [17, 21], but they cannot decide individual orbits.

The *global* view encodes a cycle by its parity word and asks whether the corresponding rational number is a positive integer. This is decided by the arithmetic of $2^l - q^k$, and it is exactly what separates $3x+1$ from $3x-1$.

The guiding question of this paper is therefore: *what determines the global dynamics of a Collatz-type map when its local predecessor structure is indistinguishable from that of another map?* Sections 2–5 give three complementary answers. The local structure of $3x+1$ and $3x-1$ is statistically identical. Cycles are characterised by an exact global divisibility condition. The basins of attraction have logarithmic densities, but their sizes differ strongly and are so far unexplained.

Contributions.

(i) Section 2 makes the limits of local methods precise. All densities of locally defined sets coincide for $3x+1$ and $3x-1$ (Proposition 2), so local branching statistics, including the cotree densities of Cayley trees, determine neither the number of cycles nor basin membership (Corollary 1). This limits proof strategies that rely only on such statistics, not proofs that combine predecessor trees with global information.

(ii) Section 3 recalls the classical correspondence between parity words and cycles (Lemma 1, due to Böhm–Sontacchi and Lagarias) in the notation of rotating binaries, and records the families of cycles obtained from binary rotations, the mirror symmetry and the product identity (10).

(iii) Section 4 takes the reverse perspective of Lagarias' rational cycles and of the shift–member duality of Belaga and Mignotte: instead of asking which cycles a given map has, it asks for which maps a given parity word yields a cycle. What is new is the exhaustive evaluation of this correspondence. One pass over all $2.2 \cdot 10^{12}$ words of length at most 40 lists all 1097 primitive cycles of all maps $3x+c$ with $|c| \le 1000$ (Proposition 4). For $0 < c < 1000$ it proves the published tables complete for all cycles with at most 40 elements. An equidistribution heuristic predicts the counts, with less scatter than a Poisson model (Figure 1, Problem 1).

(iv) Section 5 proves that every basin has a logarithmic density (Theorem 1). The proof combines the invariance of basins under first passage with the stabilisation of first passages of Gonçalves, Greenfeld and Madrid, and excludes oscillations in $\log\log y$ by comparing two windows of incommensurable ratio. Conditionally on the natural-counting transport estimate of Mazur, this becomes a natural density for $3x+1$ (Corollary 3). The section measures the basin densities of 18 maps up to 10^{10} (Figure 2). An elementary divisibility filtration (Proposition 5)

gives the densities of the gcd classes and explains the rational entries of Table 6. The values for the cycles coprime to c are stable but unexplained, and we formulate their explanation and prediction, together with the existence of natural densities, as open problems (Problem 2).

2. Local structure: one forest, several trees

The local view grows the tree of all predecessors of a cycle. For the Collatz map this tree is usually restricted to the *branching numbers*, the even numbers with an odd predecessor, and drawn as a Cayley tree with *Leftward* edges through an odd predecessor and *Upward* edges by doubling [22]. The congruence-class bookkeeping of these trees for general $an + b$ maps, including their *cotree densities*, is developed in detail by Kleinnijenhuis et al. [20]. In this section we record two facts that hold for every map of the family alike.

Let c be coprime to 6. A *branching number* is an even $n \in \mathbb{N}$ such that $(n - c)/3$ is an odd integer not divisible by 3; let S_c be the set of branching numbers. For $m \in S_c$ define the *Leftward child* $L(m) = 2^p\,(m - c)/3$ and the *Upward child* $U(m) = 2^r m$, where $p, r \geq 1$ are minimal with $L(m), U(m) \in S_c$. Both exist because 2 is a primitive root modulo 9. The following tree property of the predecessor graph is standard; we include its short proof in the present notation.

Proposition 1 (Unique parent)**.** *The maps $L, U : S_c \to S_c$ are injective, and S_c is the disjoint union of $L(S_c)$ and $U(S_c)$. In other words, every branching number has exactly one parent, either as a Leftward or as an Upward child.*

Proof. Let $n \in S_c$ and consider $n/2, n/4, \ldots$ up to the first term that is odd or lies in S_c. If this term is $n/2^r \in S_c$, then $m = n/2^r$ satisfies $U(m) = n$: the exponent r is minimal, because an earlier term in S_c would contradict the choice of r. If the first such term is an odd number $o = n/2^p$, then $3 \nmid o$, since $n \equiv c \pmod 3$ and $3 \nmid c$. Hence $m = 3o + c$ is even, $(m - c)/3 = o$ is odd and not divisible by 3, so $m \in S_c$, and $L(m) = n$ by the same minimality argument. The two cases exclude each other, and in each case the parent is determined by n. This proves surjectivity onto the disjoint union and injectivity. □

For $c = \pm 1$, the halving sequence in this proof stops after at most four steps, so whether n is a Leftward or an Upward child depends only on $n \bmod 2^4 \cdot 18 = 288$. For $3x + 1$, Kleinnijenhuis et al. count 27 Leftward and 5 Upward classes modulo 288 [20, (3.73), (3.79)], which gives the cotree density

$$d\big(L(S_c)\big) + d\big(U(S_c)\big) = \frac{27}{288} + \frac{5}{288} = \frac{2}{18} = d(S_c). \tag{5}$$

The same counts hold for $3x-1$, whose Upward classes are the negatives of those of $3x+1$ modulo 288 (Table 2). The identity (5) gives the density of the whole forest S_c, not of the tree of the trivial cycle: for $3x-1$ that tree contains only about a third of all integers (Section 5).

	$3x+1$ $(c=1)$	$3x-1$ $(c=-1)$
branching classes mod 18	$\{4, 16\}$	$\{2, 14\}$
Leftward classes mod 288	27	27
Upward classes mod 288	$\{16, 88, 160, 232, 256\}$	$\{32, 56, 128, 200, 272\}$
forest density $d(S_c)$	$27/288 + 5/288 = 2/18$	$27/288 + 5/288 = 2/18$
cycles with word length ≤ 40	1	3
basins (trees of the forest)	1 (verified for $n < 2^{71}$)	3, empirical: 32.7, 32.5, 34.8 %

TABLE 2. The mirror pair $3x \pm 1$: identical local structure (upper rows; the Upward classes are negatives of each other modulo 288), different global structure (lower rows; Sections 4 and 5).

Local indistinguishability. Let $\iota(n) = -n$. Then $T_{3,-1} = \iota \circ T_{3,1} \circ \iota$ on $\mathbb{Z}$. We call a set $A \subseteq \mathbb{N}$ *locally defined* for a map T if, for some fixed m and modulus M, membership of n in A depends only on the residues of $n, T(n), \ldots, T^m(n)$ modulo M. The branching numbers, the Leftward and Upward children and the cotree classes are locally defined, and so is, for every word B of length m, the set of n whose first m parities form B.

Proposition 2 (Local indistinguishability)**.** *Let $A \subseteq \mathbb{N}$ be locally defined for $T_{3,1}$ by residue conditions R, and let $A^- \subseteq \mathbb{N}$ be the set defined for $T_{3,-1}$ by the negated conditions $-R$. Then A and A^- are periodic sets with the same natural density.*

Proof. For $T = T_{3,\pm 1}$, the residue of $T(n)$ modulo M depends only on n modulo $2M$, hence the residue of $T^i(n)$ modulo M depends only on n modulo $2^i M$. Thus $A = \{n \in \mathbb{N} : n \bmod 2^m M \in Q\}$ for a set Q of residues, and the same condition defines a periodic set $\tilde{A} \subseteq \mathbb{Z}$ for $T_{3,1}$ on all of $\mathbb{Z}$. Since $T_{3,-1}(n) = -T_{3,1}(-n)$, we have $n \in A^-$ if and only if $-n \in \tilde{A}$, that is, $A^- = \{n \in \mathbb{N} : -n \bmod 2^m M \in Q\}$. Both sets have density $|Q|/(2^m M)$. □

Corollary 1. *A statement about $T_{3,1}$ that is formulated only in terms of the densities of locally defined sets holds if and only if the corresponding statement holds for $T_{3,-1}$. In particular, local branching statistics determine neither the number of cycles nor basin membership: $3x+1$ has one known cycle and $3x-1$ has three (Section 4), and Table 3 shows that membership in a basin is not a function of any residue class.*

Remark 1 (Scope). Corollary 1 limits proof strategies that use only local branching statistics, such as cotree densities. It does not exclude proofs that combine the predecessor tree with global information that differs between $3x+1$ and $3x-1$: the size of the numbers involved, as in the descent $(x+1)/2 < x$ for the map $x+1$, or the arithmetic of $2^l - 3^k$ (Section 3).

Residue classes do not decide membership. Which tree of the forest a branching number belongs to is not a property of its residue class, for none of the moduli 18, 288 and 4608 used in this kind of bookkeeping. Table 3 shows this for $3x-1$, where all trees are known: members of the same class lie in different trees, in both directions. A check of the representatives $\ell = 0$ can therefore refute a conjecture (the number 14 refutes it for $3x-1$), but it cannot confirm one. For $3x+1$, the representatives 4 and 16 lie in the tree of the root; whether all members of $18\ell+4$ and $18\ell+16$ do is the Collatz conjecture itself.

class	members in the tree of 2	first member outside	its cycle
$18\ell+14$	32 ($\ell=1$)	14 ($\ell=0$)	through 5
$18\ell+2$	2 ($\ell=0$, the root)	20 ($\ell=1$)	through 5
$288\ell+2$	$2, 290, 578, 866$ ($\ell \le 3$)	1154 ($\ell=4$)	through 17
$4608\ell+2$	2 ($\ell=0$)	4610 ($\ell=1$)	through 17

TABLE 3. Members of the same residue class in different trees of the forest of $3x-1$. For example, $32 \to 16 \to 8 \to 4 \to 2$, whereas $14 \to 7 \to 20 \to 10 \to 5 \to 14$.

3. Parity words and cycles

Fix odd $q \ge 1$ and odd c, and write $T = T_{q,c}$. A *word* $B = b_0 b_1 \cdots b_{l-1} \in \{0,1\}^l$ prescribes the parities of l consecutive iterates. Its *type* is (l,k) with $k = b_0 + \cdots + b_{l-1}$, the number of odd steps; we write

$$D = D(l,k) = 2^l - q^k. \tag{6}$$

Let $0 \le p_1 < p_2 < \cdots < p_k \le l-1$ be the positions of the ones in B; the position 0 refers to the first step. The rotating-binary function of Cox et al. [23], adapted from Halbeisen and Hungerbühler [24], is

$$z_q(B) = \sum_{j=1}^{k} q^{\,k-j}\, 2^{p_j}. \tag{7}$$

Let $\sigma B = b_1 \cdots b_{l-1} b_0$ denote the left rotation of B, and let $\mathbb{Z}_{(2)}$ be the ring of rationals with odd denominator. The following correspondence is classical [9, 25]; we state it in the notation of rotating binaries and sketch the proof.

Lemma 1 (Cycles from words; Böhm–Sontacchi, Lagarias)**.** *Let B be a word of type (l,k) with $D \neq 0$, and put*

$$x_B = \frac{c\, z_q(B)}{D} \in \mathbb{Z}_{(2)}.$$

Then x_B is a periodic point of T (extended to $\mathbb{Z}_{(2)}$) whose parity sequence is $BBB\cdots$, and it is the only such point in $\mathbb{Z}_{(2)}$. The orbit of x_B is $x_B, x_{\sigma B}, x_{\sigma^2 B}, \ldots$. Consequently:

(a) T has an integer cycle with parity word B if and only if $D \mid c\, z_q(B)$;
(b) for $k \geq 1$ this cycle consists of positive integers if and only if $c\, D > 0$.

Sketch of proof. Let $T_0(x) = x/2$, $T_1(x) = (qx+c)/2$ and $T_B = T_{b_{l-1}} \circ \cdots \circ T_{b_0}$. By induction on l, $T_B(x) = (q^k x + c\, z_q(B))/2^l$, whose unique fixed point is x_B; it lies in $\mathbb{Z}_{(2)}$ because D is odd. From $T_{\sigma B} \circ T_{b_0} = T_{b_0} \circ T_B$ we get $T_{b_0}(x_B) = x_{\sigma B}$, and $2x_{\sigma B} = q^{b_0} x_B + b_0 c$ shows that x_B has parity b_0. Hence the orbit of x_B follows B periodically. If $y \in \mathbb{Z}_{(2)}$ also has parity sequence $BBB\cdots$, then $T^{jl}(y) - T^{jl}(x_B) = (q^k/2^l)^j (y - x_B)$ lies in $\mathbb{Z}_{(2)}$ for all j, so $y = x_B$. Statements (a) and (b) follow because $z_q(B) > 0$ for $k \geq 1$. □

Every positive integer cycle of T arises in this way from the word of its parities, and its members are obtained by rotating the word. This is the precise sense in which cycles are *rotating binaries*. The reverse direction, asking for which maps a given word yields a cycle, is also classical: it is Lagarias' theory of rational cycles [9] and the shift–member duality of Belaga and Mignotte [14, Theorem 4].

Example 1. Take $q = 5$ and the word $B = 111110000000$ of type $(12, 5)$. Then $D = 2^{12} - 5^5 = 971$. With $c = 971$ we get $x_B = z_5(B) = 5^4 + 5^3 \cdot 2 + 5^2 \cdot 2^2 + 5 \cdot 2^3 + 2^4 = 1031$, and the twelve rotations of B produce the cycle

$$(1031, 3063, 8143, 20843, 52593, 131968, 65984, 32992, 16496, 8248, 4124, 2062)$$

of $5x + 971$. Its odd members are $1031, 3063, 8143, 20843, 52593$. The binary representation of B reflects the course of the cycle: the ones mark the odd members.

Words with $D = c$. If $c = 2^l - q^k > 0$, then $D = c$, and Lemma 1 gives

$$x_B = z_q(B) \qquad \text{for every word } B \text{ of type } (l,k). \tag{8}$$

Every such word therefore yields a positive integer cycle of $T_{q,c}$, with members $z_q(\sigma^i B)$, and each necklace (rotation class) of type (l,k) gives one cycle. This explains the cycles of Table 4, where $7 = 2^5 - 5^2$, $103 = 2^7 - 5^2$, $3 = 2^7 - 5^3$ and $131 = 2^8 - 5^3$, and the cycle families of $3x + 5$ ($5 = 2^5 - 3^3$) and $3x + 13$ ($13 = 2^8 - 3^5$) in Section 4.

map	B_{min}	B_{max}	r	c	cycle
$5x+7$	00101	10100	2	2^5-5^2	$9, 26, 13, 36, 18$
$5x+103$	0001001	1001000	3	2^7-5^2	$13, 84, 42, 21, 104, 52, 26$
$5x+3$	0010101	1010100	2	2^7-5^3	$61, 154, 77, 194, 97, 244, 122$
$5x+131$	00100101	10100100	5	2^8-5^3	$77, 258, 129, 388, 194, 97, 308, 154$

TABLE 4. Cycles determined via binary rotation [26]; r is the left-rotational distance from $B_{\min}$ to $B_{\max}$, and $c = 2^l - 5^k$, see (8). The notebook is available on GitHub [27].

Mirror and $D = 1$. The conjugation $x \mapsto -x$ gives

$$T_{q,-c}(x) = -T_{q,c}(-x), \tag{9}$$

so the cycles of $3x-1$ are the negatives of the cycles of $3x+1$ on the negative integers. By Lemma 1(b) they come from words with $D < 0$: the three cycles of $3x-1$ through 1, 5 and 17 have the words 1, 110 and 11110111000, with $D = -1$, -1 and -139 (since $2^3 < 3^2$ and $2^{11} < 3^7$). If $q = 2^m - 1$, the word 10^{m-1} has $D = 2^m - q = 1$ and $z_q = 1$, so $x_B = c$; for $c = 1$ the number 1 lies on a cycle. This is the power-of-two root $c = q + 1 = 2^m$ of the family $a = 2^m - 1$ studied in [20].

Remark 2 (The case $q = 1$). For $q = 1$ we have $z_1(B) = \sum_j 2^{p_j}$, the integer whose binary digits, least significant first, are $b_0, \ldots, b_{l-1}$. For $c = 2^l - 1$, the map $T_{1,c}$ acts on $\{0, \ldots, 2^l - 1\}$ as the cyclic right rotation of l-bit numbers. For example, the cycles of $x + 7$ are $\{1, 4, 2\}$, $\{3, 5, 6\}$ and $\{7\}$, the rotation classes of 001, 011 and 111. For $c = 1$, by contrast, $0 < z_1(B) < 2^l - 1$ unless $B = 1^l$, so $T_{1,1}$ has only the cycle $\{1\}$. This also follows from the descent $(x + 1)/2 < x$ for odd $x > 1$, a global, Archimedean argument. This is the trivial base case $1n + 1$ of the family $an + b$.

The product identity. Let $a_1, \ldots, a_k$ be the odd members of a positive cycle of (1) of type (l, k), in cyclic order, and put $A_i = qa_i$. Dividing (4) by q^k gives

$$\prod_{i=1}^{k} \Big(1 + \frac{1}{A_i}\Big) = \frac{2^l}{q^k},$$

and for odd $q > 1$ the right-hand side determines (l, k). Hence the members of two cycles satisfy the product equality

$$(A_1 + 1) \cdots (A_m + 1) \cdot B_1 \cdots B_n = A_1 \cdots A_m \cdot (B_1 + 1) \cdots (B_n + 1) \tag{10}$$

if and only if the cycles have the same type. For example, the two non-trivial cycles of $5x+1$ both have type $(7, 3)$, so (10) holds for $A = 5 \cdot (13, 33, 83)$ and $B = 5 \cdot (27, 17, 43)$; both products equal $2^7/5^3$. Likewise, both cycles of $181x+1$ have type $(15, 2)$ (Table 5).

4. The decoded zoo: cycles live where $2^l \approx q^k$

By Lemma 1, the positive integer cycles of $T_{q,c}$ with word length $l \le L$ can be found exhaustively. One enumerates all words B of length $l \le L$ with $c\,D > 0$ and tests whether $D \mid c\,z_q(B)$. For $3x+c$ with $c > 0$ and $L = 30$ these are $1.96 \cdot 10^9$ words, which takes a few minutes on a single core. The program and its outputs are available in [28].

Proposition 3 (Census)**.** *The cycles listed in Table 5 are all positive integer cycles of $T_{q,c}$ whose parity word has length at most L, with $L = 40$ for $3x+1$, $3x-1$, $3x+5$, $3x+13$ (by Proposition 4), $L = 28$ for $5x+1$ and $L = 40$ for $181x+1$.*

map	cycle minima	(l,k)	$D = 2^l - q^k$	$N_{l,k}$	E
$3x+1$	1	$(2,1)$	1	1	1.00
$3x-1$	1	$(1,1)$	-1	1	1.00
	5	$(3,2)$	-1	1	1.00
	17	$(11,7)$	-139	30	0.22
$3x+5$	5	$(2,1)$	1	1	1.00
	1	$(3,1)$	5	1	1.00
	19, 23	$(5,3)$	5	2	2.00
	187, 347	$(27,17)$	5 077 565	312 455	0.31
$3x+13$	13	$(2,1)$	1	1	1.00
	1	$(4,1)$	13	1	1.00
	211, 227, 251, 259, 283, 287, 319	$(8,5)$	13	7	7.00
	131	$(24,15)$	2 428 309	54 484	0.29
$5x+1$	1	$(5,2)$	7	2	0.29
	13, 17	$(7,3)$	3	5	1.67
$181x+1$	27, 35	$(15,2)$	7	7	1.00

Table 5. All positive integer cycles of $T_{q,c}$ with word length $l \le L$ (Proposition 3), given by their minima and decoded by their type (l,k). $N_{l,k}$ is the number of binary necklaces of type (l,k), and $E = N_{l,k}\gcd(c,D)/|D|$ is the heuristic expected number of cycles of that type if $c\,z_q(B)$ were equidistributed modulo D.

Near-coincidences. In every row of Table 5 with a non-trivial cycle, $|D|$ is small compared with 2^l. For example, $2^{15} - 181^2 = 32768 - 32761 = 7$, $2^7 - 5^3 = 128 - 125 = 3$ and $|D|/2^l = 0.038$ for the type $(27,17)$ of $3x+5$. Equivalently, l/k is a good rational

approximation of $\log_2 q$:

$$\begin{aligned} 15/2 &= 7.5 && \approx \log_2 181 = 7.49985, \\ 7/3 &= 2.333 && \approx \log_2 5 = 2.32193, \\ 27/17,\ 8/5,\ 11/7 &= 1.588,\ 1.600,\ 1.571 && \approx \log_2 3 = 1.58496. \end{aligned}$$

For $3x-1$ the approximations come from below ($2^l < 3^k$), in accordance with (9). The heuristic count E explains the pattern. A small $|D|$ makes the divisibility $D \mid c\, z_q(B)$ likely, while a large $|D|$ needs an astronomical number of words.

The near miss of $3x+1$. The type $(8,5)$ illustrates where the $3x+1$ problem is decided.

Corollary 2. *The $3x+1$ map has no cycle of type $(8,5)$, and the seven cycles of $3x+13$ of this type are exactly the seven candidates.*

Proof. By Lemma 1, a $3x+1$ cycle of type $(8,5)$ requires $13 = 2^8 - 3^5$ to divide $z_3(B)$ for a word B of that type. For $c = 13$ we have $D = c$, so by (8) the values $z_3(B)$ are exactly the members of the seven cycles of $3x+13$ of type $(8,5)$. Their minima $211, 227, 251, 259, 283, 287, 319$ are congruent to $3, 6, 4, 12, 10, 1, 7$ modulo 13. Since $T_{3,13}$ maps non-multiples of 13 to non-multiples of 13, no member of these cycles is divisible by 13. □

The heuristic expectation for this type is $E = 7/13 \approx 0.54$; the actual count is 0. Belaga and Mignotte call $2^l - 3^k$ a Collatz number and the values $z_3(B)$ of the words of type (l,k) that begin with an odd step its corona. In their language, Corollary 2 states that the Collatz number $2^8 - 3^5 = 13$ divides no member of its corona; they showed that the absence of non-trivial positive $3x+1$ cycles is equivalent to this property for every non-trivial type [14, Theorem 5]. For longer types, the methods of Steiner, Simons–de Weger and Hercher [3–5] combine lower bounds for $|l \log 2 - k \log 3|$ from Baker's theory with the verification limit. These bounds show that D is too large for the divisibility to occur in cycles with few runs. Whatever proves the absence of cycles must see this arithmetic. The local structure of Section 2 does not see it, because it is identical for $3x+1$ and $3x+13$ up to scaling.

All maps at once: rational cycles. We now take the reverse perspective: instead of asking which cycles a given map has, we ask for which maps a given parity word yields a cycle. This is the point of view of Lagarias' rational cycles [9] and of the shift–member duality of Belaga and Mignotte [14, Section 7]. Lemma 1 answers the question for every word. Let d_0 be the denominator of $z_3(B)/D$ in lowest terms. Then $x_B = c\, z_3(B)/D$ is a positive integer if and only if $c = \operatorname{sign}(D)\, m\, d_0$ with m odd, and the cycle is m times the *primitive* cycle of $3x + c_0$, $c_0 = \operatorname{sign}(D)\, d_0$. These are Lagarias' rational

cycles of $3x+1$ with denominator d_0 [9]. A single pass over all words therefore lists the primitive cycles of all maps with $|c_0| \le C$. The denominator comes from the continued-fraction expansion of $z_3(B)/|D|$, which can be stopped as soon as a convergent has a denominator larger than C.

Proposition 4 (Rational cycles)**.** *For word length $l \le 40$ there are exactly* 1097 *primitive cycles of the maps $3x + c_0$ with $|c_0| \le 1000$: 763 with $c_0 > 0$ and 334 with $c_0 < 0$, for* 199 *distinct values of $|c_0|$. Every cycle of a map $3x + c$ with $|c| \le 1000$ and word length at most* 40 *is an odd multiple of one of them.*

The computation checked all 2 199 023 255 510 words in 512 shards of about six minutes each on the HPC cluster of the EDIH Hamburg, with a parity check of every reported cycle. The program was validated against an independent implementation in exact rational arithmetic (for $l \le 16$ and $|c_0| \le 50$) and against the single-map census for $3x \pm 1$, $3x + 5$ and $3x + 13$.

Comparison with the published tables. For each of the 333 values $0 < k < 1000$ with $\gcd(k, 6) = 1$, the primitive cycles of $3x + k$ listed by Belaga, Mignotte and Morley [11, 12] that have at most 40 elements coincide exactly with those of Proposition 4; there are 763 of them. The other 462 listed cycles have between 41 and 426 elements, and for 153 of these maps every known primitive cycle is longer than 40. Proposition 4 therefore turns the conjecturally complete tables into a theorem for all cycles with at most 40 elements and $k < 1000$.

Heuristic. If $z_3(B)$ were equidistributed modulo $|D|$, the reduced denominator of $z_3(B)/|D|$ would equal a divisor e of $|D|$ with probability $\varphi(e)/|D|$. A primitive cycle corresponds to one aperiodic necklace of words, so the expected number of primitive cycles of type (l, k) with $|c_0| \le C$ is

$$E_C(l,k) = L(l,k)\,\frac{S_C(D)}{|D|}, \qquad S_C(D) = \sum_{e \,|\, |D|,\ e \le C} \varphi(e),$$

where $L(l, k)$ is the number of aperiodic binary necklaces of type (l, k). If all divisors of $|D|$ are at most C, then $S_C(D) = |D|$, every necklace yields a cycle, and E_C is exact; for example, $D(10, 5) = 781 = 11 \cdot 71$ gives all 25 necklaces. Summed over $l \le 40$, the heuristic predicts 1148.7 primitive cycles against 1097 observed, and it reproduces the counts per word length including the peaks (Figure 1). The agreement has a deterministic and a statistical part. For the 44 types with $|D| \le 1000$, all divisors of $|D|$ are admissible, every necklace yields a cycle, and the 227 cycles of these types are predicted exactly. For the other 116 types with $E_{1000} \ge 1$, the heuristic predicts 879.8 cycles and we observe 839; the remaining types with $E_{1000} < 1$ contribute 41.9 expected and 31 observed cycles. For the 116 types, $\sum (O - E)^2/E = 71.4$, a dispersion index

of 0.62, nearly constant across the ranges $1 \le E < 3$, $3 \le E < 10$ and $E \ge 10$. The counts thus scatter less than a Poisson model allows. This suggests that the residues $z_3(B) \bmod |D|$ of the words of a type are distributed more evenly than independent random residues would be.

Problem 1. *Why does the equidistribution heuristic predict the number of primitive cycles so well, and why are the counts underdispersed? Can the distribution of $z_3(B)$ modulo $|D|$ over the words of a type be controlled?*

Longer words. The census is complete for $l \le 40$ only. A search over the starting values $n < 10^5$ of the maps of Section 5 finds three further cycles: the cycle of $3x+23$ through 41 ($l = 43$) and two cycles of $3x+29$ through 3811 and 7055 ($l = 65$, $k = 41$). The last two sit exactly at the convergent $65/41$ of $\log_2 3$, the one after $19/12$.

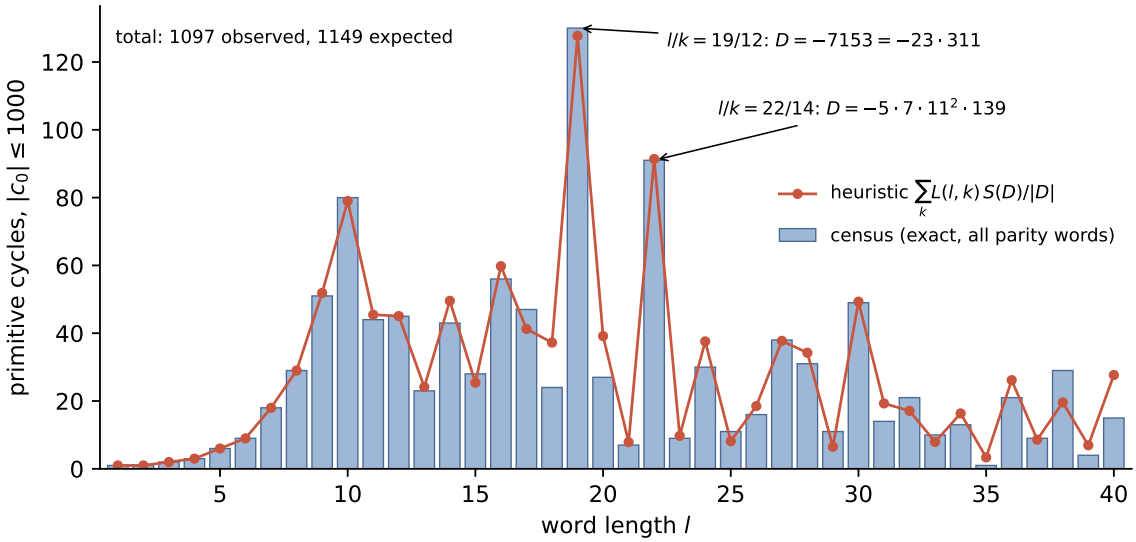


FIGURE 1. Primitive cycles of all maps $3x+c_0$ with $|c_0| \le 1000$ per word length l: census (bars) and heuristic $\sum_k E_{1000}(l,k)$ (line). The peaks at $l = 19$ and $l = 22$ come from $D(19,12) = -7153 = -23 \cdot 311$ and $D(22,14) = -5 \cdot 7 \cdot 11^2 \cdot 139$.

5. MEASURING FORESTS: BASIN DENSITIES

For a cycle $\mathcal{C}$ of $T_{q,c}$, its *basin* is $\beta(\mathcal{C}) = \{n \in \mathbb{N} : T^j(n) \in \mathcal{C} \text{ for some } j\}$, and we write $d_N(\mathcal{C}) = |\beta(\mathcal{C}) \cap [1,N]|/N$. The basins are the trees of the forest of Section 2; the local structure fixes only the total density of all trees. We report empirical shares d_N.

Except for the basins covered by Proposition 5, it is not known whether the natural density $\lim_{N\to\infty} d_N$ exists; Theorem 1 below shows that the logarithmic density always does.

Two entries of Table 6 are rational, $0.2000 = 1/5$ and $0.0769 \approx 1/13$. They are explained by the following elementary gcd invariance, which is implicit in Lagarias' primitive cycles [9] and explicit in [14, (19)].

Proposition 5 (Divisibility filtration)**.** *Let* $\gcd(c,6)=1$ *and, for a divisor* e *of* $|c|$*, let* $B_e=\{n\ge 1 : \gcd(n,c)=e\}$*. Then* $T_{3,c}(B_e)\subseteq B_e$*, the map* $n\mapsto n/e$ *conjugates* $T_{3,c}$ *on* B_e *to* $T_{3,c/e}$ *on the integers coprime to* c/e*, and* B_e *has natural density* $\varphi(|c|/e)/|c|$*.*

Proof. Every prime divisor p of c satisfies $p\ge 5$. Since $p \mid (3n+c)/2$ if and only if $p \mid n$, and $p \mid n/2$ if and only if $p \mid n$, the map preserves $\gcd(n,c)$. For odd e, the numbers em and m have the same parity, and $T_{3,c}(em) = e\,T_{3,c/e}(m)$. Finally, $n=em$ lies in B_e if and only if $\gcd(m,c/e)=1$, a condition of density $\varphi(|c|/e)/(|c|/e)$ for m, hence of density $\varphi(|c|/e)/|c|$ for n. □

map	cycle (minimum)	d_{10^6}	d_{10^8}	$d_{10^{10}}$
$3x-1$	1	0.3277	0.3270	0.3270
	5	0.3234	0.3247	0.3250
	17	0.3490	0.3483	0.3480
$3x+5$	1	0.1410	0.1404	0.1403
	5	0.2000	0.2000	0.2000
	19	0.4964	0.4979	0.4979
	23	0.0930	0.0922	0.0922
	187	0.0325	0.0326	0.0325
	347	0.0371	0.0369	0.0370
$3x+13$	1	0.4769	0.4775	0.4775
	13	0.0769	0.0769	0.0769
	131	0.2522	0.2517	0.2517
	211	0.0334	0.0334	0.0334
	227	0.0191	0.0189	0.0189
	251	0.0197	0.0199	0.0198
	259	0.0388	0.0386	0.0386
	283	0.0254	0.0253	0.0253
	287	0.0430	0.0432	0.0433
	319	0.0146	0.0145	0.0145

TABLE 6. Empirical basin densities d_N of all cycles of $3x-1$, $3x+5$ and $3x+13$ (every $n \le 10^{10}$ lies in one of the listed basins).

Consequently, for every cycle $\mathcal{C}$ of $3x+c/e$ coprime to c/e, the basin of $e\,\mathcal{C}$ in $3x+c$ has exactly $1/e$ times the density of the basin of $\mathcal{C}$ in $3x+c/e$, whenever the latter exists. Only the cycles coprime to c are new in each map. Proposition 5 thus determines the densities of the gcd classes B_e; the density of an individual basin follows only in special cases. If B_1 contains a single cycle, its basin has density $\varphi(|c|)/|c|$, provided all orbits reach known cycles. The basins of the cycles through 5 in $3x+5$ and through 13 in $3x+13$ are 5 and 13 times the basin of 1 in $3x+1$, so their densities are $1/5$ and $1/13$ provided that almost all integers reach 1 under $3x+1$.

Proposition 6 (Verification)**.** *For* $3x-1$ *and the* 17 *maps* $3x+c$ *with* $0<c<50$ *and* $\gcd(c,6)=1$*, every orbit of* $1 \le n \le 10^{10}$ *reaches one of the cycles of Proposition 4 or one of the three cycles with longer words found above.*

Figure 2 shows the resulting split of the forest for $3x-1$ and the 17 maps $3x+c$ with $0<c<50$ and $\gcd(c,6)=1$. In the left panel, coloured segments are the basins of

cycles coprime to c, labelled by their minima where space permits; grey segments are the basins of cycles of multiples of a divisor $e > 1$ of c, whose densities are $1/e$ times basin densities of $3x + c/e$ by Proposition 5. For $c = 7$, 19, 31, 41 and 43 the integers coprime to c form a single basin of density $(c-1)/c$. The grey parts of $3x+25$, $3x+35$ and $3x + 49$ reproduce the densities of $3x + 5$ and $3x + 7$, scaled by $1/5$ and $1/7$, to all printed digits. The densities are stable. The right panel follows the three basins of $3x - 1$ through the dyadic windows $[2^j, 2^{j+1})$, and in the window $[2^{32}, 2^{33})$ the densities of all basins differ from those up to 10^{10} by at most $2.4 \cdot 10^{-4}$. Within the cycles coprime to c, however, no pattern is visible. In $3x+5$, the cycle through 19 attracts almost half of all integers, while the cycle through 23 of the same type $(5, 3)$ attracts less than a tenth. In $3x - 1$, by contrast, the three basins are nearly equal, as already observed in [18, Remark 1.7]. The type of a cycle, its length and number of odd steps, therefore does not determine its attractiveness.

Existence of the densities. For $A \subseteq \mathbb{N}$, the *logarithmic density* is

$$\delta(A) = \lim_{N\to\infty} \frac{1}{\log N} \sum_{\substack{n<N \\ n\in A}} \frac{1}{n}, \tag{11}$$

if the limit exists. If the natural density of A exists, then $\delta(A)$ exists and has the same value; the converse fails in general. Tao proved his theorem on almost bounded orbits in logarithmic density [17], and Gonçalves, Greenfeld and Madrid extended his method to a class of maps that contains $T_{3,c}$ [18]. They observed numerically that the predecessor set of an integer a appears to have a natural density $\kappa(a)$, and that each of the three cycles of $3x - 1$ attracts about a third of all integers; they also proposed a formulation with logarithmically distributed starting values [18, Remark 1.7]. In this logarithmic form, existence follows from their stabilisation of first passages, because a basin is not changed by first passage. For natural densities see Corollary 3 and Remark 6.

We call $A \subseteq \mathbb{N}$ *completely invariant* under T if $n \in A \iff T(n) \in A$. Every basin is completely invariant, and so is the set of starting values with unbounded orbits.

Theorem 1 (Logarithmic basin densities)**.** *Let $c \geq -1$ be odd, and let $A \subseteq \mathbb{N}$ be completely invariant under $T_{3,c}$. Then $\delta(A)$ exists. In particular, the basin of every cycle $\mathcal{C}$ of $T_{3,c}$ has a logarithmic density $\delta(\mathcal{C})$, and if $\lim_{N\to\infty} d_N(\mathcal{C})$ exists, it equals $\delta(\mathcal{C})$.*

For the proof we use the Syracuse form of $T_{3,c}$ on the odd integers, $\mathrm{Syr}(m) = (3m + c)/2^{\nu_2(3m+c)}$, where ν_2 is the 2-adic valuation. For $x \geq 1$ and odd m, the *first passage time* below x is $\tau_x(m) = \inf\{j \geq 0 : \mathrm{Syr}^j(m) \leq x\}$, and the *first passage* is $\mathrm{Pass}_x(m) = \mathrm{Syr}^{\tau_x(m)}(m)$, with $\mathrm{Pass}_x(m) = 1$ if $\tau_x(m) = \infty$. Let $\mu(I) = \sum_{m\in I,\ m \text{ odd}} 1/m$, so that

$\mu([u,v)) = \frac{1}{2}\log(v/u) + O(1/u)$. For $1 \le u < v$ let $\mathbf{Log}_{u,v}$ be a random odd integer in $[u,v)$ with $\mathbb{P}(\mathbf{Log}_{u,v} = m) = 1/(m\,\mu([u,v)))$, and write d_{TV} for the total variation distance. The map $n \mapsto n/2$ (n even), $n \mapsto 3n + c$ (n odd) belongs to the class of [18, Definition 1.2] with $p = 2$, $q = 3$ and $r(1) = c$, and its Syracuse map is Syr; the condition $c \ge -1$ makes it map $\mathbb{N}$ into itself. It satisfies the hypotheses (a)–(c) of [18, Theorem 1.3]: $\gcd(2,3) = 1$, $3 < 2^2$, and $3 + c$ is even. These hypotheses impose no condition on $\gcd(3,c)$. The proof of [18, Proposition 3.5], however, is carried out after the normalisation $\gcd(q, r(1)) = 1$ of [18, Section 2.2], that is, for $3 \nmid c$; Step 0 below reduces Theorem 1 to this case.

Proof of Theorem 1. Step 0 (factors 3 in c). Suppose $3 \mid c$, and write $c = 3c'$ with $c' \ge 1$ odd. Then $T_{3,c}(3m) = 3\,T_{3,c'}(m)$ for all m, and $T_{3,c}(n) = 3(n + c')/2$ for odd n. Hence $A' = \{m : 3m \in A\}$ is completely invariant under $T_{3,c'}$, and an odd n lies in A if and only if $(n + c')/2 \in A'$. Since

$$\sum_{\substack{n<Y,\ n \text{ odd} \\ (n+c')/2 \in A'}} \frac{1}{n} = \frac{1}{2} \sum_{\substack{u<Y/2 \\ u \in A'}} \frac{1}{u} + O(1),$$

Step 1 below shows that $\delta(A)$ exists and equals $\delta(A')$ whenever $\delta(A')$ exists. Repeating this step, we may assume $3 \nmid c$.

Input. For odd $c \ge -1$ with $3 \nmid c$, the stabilisation of first passages of Gonçalves, Greenfeld and Madrid [18, Proposition 3.5], the analogue of [17, Proposition 1.11], provides $\varepsilon_0 \in (0,1)$ such that for all $\alpha, \lambda \in (1, 1+\varepsilon_0)$ and all $x \ge 2$

$$\mathbb{P}\big(\tau_x(\mathbf{Log}_{x^\lambda, x^{\alpha\lambda}}) = \infty\big) \ll x^{-\varepsilon_0}, \tag{12}$$

$$d_{\mathrm{TV}}\big(\mathrm{Pass}_x(\mathbf{Log}_{x^\lambda, x^{\alpha\lambda}}), \mathrm{Pass}_x(\mathbf{Log}_{x^{\alpha\lambda}, x^{\alpha^2\lambda}})\big) \ll (\log x)^{-\varepsilon_0}, \tag{13}$$

with implied constants that do not depend on x. Choose $\varepsilon_1 > 0$ with $(1+\varepsilon_1)^2 < 1 + \varepsilon_0$ and fix $\lambda_0 \in (1, 1+\varepsilon_1)$.

Step 1 (odd numbers). Write $n = 2^k m$ with m odd. Since $T_{3,c}^k(n) = m$, we have $n \in A$ if and only if $m \in A$. The odd part $A_1 = A \cap (2\mathbb{N} + 1)$ satisfies $m \in A_1 \iff \mathrm{Syr}(m) \in A_1$. Let $\kappa(Y) = \mu(A_1 \cap [1,Y))/\mu([1,Y))$. If $\kappa(Y) \to \delta$, then

$$\sum_{\substack{n<N \\ n \in A}} \frac{1}{n} = \sum_{k \ge 0} 2^{-k} \mu\big(A_1 \cap [1, N/2^k)\big) = \delta \log N + o(\log N),$$

so $\delta(A) = \delta$. It remains to show that $\kappa(Y)$ converges.

Step 2 (invariance under first passage). If $m > x$ and $\tau_x(m) < \infty$, then $m \in A_1 \iff \mathrm{Pass}_x(m) \in A_1$. Hence, for every random odd integer $\mathbf{m} > x$,

$$\big|\mathbb{P}(\mathbf{m} \in A_1) - \mathbb{P}\big(\mathrm{Pass}_x(\mathbf{m}) \in A_1\big)\big| \le \mathbb{P}\big(\tau_x(\mathbf{m}) = \infty\big). \tag{14}$$

Step 3 (adjacent windows). Let $\alpha \in (1, 1+\varepsilon_1)$ and $F_\alpha(y) = \mathbb{P}(\mathbf{Log}_{y,y^\alpha} \in A_1) = \mu(A_1 \cap [y, y^\alpha))/\mu([y, y^\alpha))$. For $y \geq 4$ put $x = y^{1/\lambda_0}$. The windows $[y, y^\alpha) = [x^{\lambda_0}, x^{\alpha\lambda_0})$ and $[y^\alpha, y^{\alpha^2}) = [x^{\alpha\lambda_0}, x^{\alpha^2\lambda_0})$ lie above x, and $\alpha\lambda_0 < 1+\varepsilon_0$. By (14) for both windows, (12) with $\lambda = \lambda_0$ and $\lambda = \alpha\lambda_0$, and (13) with $\lambda = \lambda_0$,

$$|F_\alpha(y) - F_\alpha(y^\alpha)| \ll x^{-\varepsilon_0} + (\log x)^{-\varepsilon_0} \ll (\log y)^{-\varepsilon_0}.$$

Step 4 (a limit that is periodic in $\log\log y$*).* Put $t = \log\log y$, $a = \log\alpha$ and $f_\alpha(t) = F_\alpha(\exp e^t)$. Then $f_\alpha(t+a) = F_\alpha(y^\alpha)$, so $|f_\alpha(t) - f_\alpha(t+a)| \ll e^{-\varepsilon_0 t}$. These differences are summable along $t, t+a, t+2a, \ldots$, so $\psi_\alpha(t) = \lim_{k\to\infty} f_\alpha(t+ka)$ exists, $\psi_\alpha(t+a) = \psi_\alpha(t)$, and $|f_\alpha(t) - \psi_\alpha(t)| \ll e^{-\varepsilon_0 t}$. For $W = [y, y^\alpha)$, $W' = [y^{1+\eta}, y^{\alpha(1+\eta)})$ and $0 < \eta \leq 1$,

$$|F_\alpha(y) - F_\alpha(y^{1+\eta})| \leq \frac{2\mu(W \triangle W')}{\mu(W)} \leq \frac{2(1+\alpha)\eta}{\alpha - 1} + O\Big(\frac{1}{\log y}\Big).$$

Since the shift in t is $\log(1+\eta)$, this gives $|f_\alpha(t) - f_\alpha(t')| \ll_\alpha |t - t'| + e^{-t}$ for $t \leq t' \leq t + \log 2$, so ψ_α is Lipschitz continuous.

Step 5 (from windows to κ*).* Let $\tilde\kappa(t) = \kappa(\exp e^t)$ and $h(t) = \mu([1, \exp e^t)) = \frac{1}{2}e^t + O(1)$. The identity $\mu(A_1 \cap [y, y^\alpha)) = F_\alpha(y)\,\mu([y, y^\alpha))$ reads $\tilde\kappa(t+a)h(t+a) = \tilde\kappa(t)h(t) + f_\alpha(t)\big(h(t+a) - h(t)\big)$, hence

$$\tilde\kappa(t+a) = e^{-a}\tilde\kappa(t) + (1 - e^{-a})f_\alpha(t) + O(e^{-t}).$$

For fixed t, the numbers $D_k = \tilde\kappa(t+ka) - \psi_\alpha(t)$ satisfy $|D_{k+1}| \leq e^{-a}|D_k| + O(e^{-\varepsilon_0(t+ka)})$, because $\psi_\alpha(t+ka) = \psi_\alpha(t)$ and $\varepsilon_0 < 1$. Thus $D_k \to 0$ uniformly for t in compact intervals, and $\tilde\kappa(t) - \psi_\alpha(t) \to 0$ as $t \to \infty$.

Step 6 (two incommensurable windows). Let $\alpha' = \alpha^{1/\sqrt{2}}$ and $a' = \log\alpha'$. Steps 3–5 for α' give a continuous a'-periodic function $\psi_{\alpha'}$ with $\tilde\kappa(t) - \psi_{\alpha'}(t) \to 0$, so $\psi_\alpha(t) - \psi_{\alpha'}(t) \to 0$. Fix t. As $\psi_\alpha(t+ka) = \psi_\alpha(t)$, we get $\psi_{\alpha'}(t+ka) \to \psi_\alpha(t)$ for $k \to \infty$. Since $a/a' = \sqrt{2}$ is irrational, the points $t + ka$ with $k \geq k_0$ are dense modulo a' for every k_0, and $\psi_{\alpha'}$ is continuous; hence $\psi_{\alpha'}$ is constant, equal to $\psi_\alpha(t)$. As t was arbitrary, ψ_α and $\psi_{\alpha'}$ are the same constant δ, and $\kappa(Y) \to \delta$ by Step 5.

Finally, if the natural density of A exists, partial summation shows that the logarithmic density exists with the same value. □

Remark 3 (Consequences and limits). For $c = 1$, Theorem 1 shows that the integers whose orbit reaches 1 form a set with a logarithmic density; the Collatz conjecture asserts that this set is $\mathbb{N}$. We have not found this consequence stated in the literature; Corollary 3 upgrades it to natural density, conditionally on Mazur's transport estimate. The set of starting values with unbounded orbits is completely invariant as well, so it also has a logarithmic density. Existence is shared by all maps of the family, in particular by $3x+1$ and $3x-1$; what separates them are the values $\delta(\mathcal{C})$. The method does not

reach natural densities, because the stabilisation of first passages controls logarithmic averages only. Maps $3x + c$ with $c \leq -3$ do not map $\mathbb{N}$ into itself and are not covered; by (9) they are the maps $3x + |c|$ on the negative integers.

Remark 4 (Predecessor sets). Step 2 uses only that membership is preserved by the first passage below x for all large x. This also holds for the set of n whose orbit contains a fixed odd integer a, once $x \geq a$: an orbit that starts above x stays above x until its first passage, so it cannot meet a before. Step 1 applies as well, because an orbit can meet the odd number a only after its initial halvings. Hence these sets have logarithmic densities, which answers the existence question of [18, Remark 1.7] in logarithmic form for every odd a. Its natural-density form remains open, except for $c = 1$ under the hypothesis of Corollary 3, whose proof applies to these sets as well. For a on a cycle, the set is the basin of that cycle.

Remark 5 (A first-passage formula). For $3 \nmid c$, Steps 2, 4 and 6 and (12) give, for all $\alpha, \lambda \in (1, 1 + \varepsilon_1)$,

$$\delta(\mathcal{C}) = \lim_{x\to\infty} \mathbb{P}\big(\mathrm{Pass}_x(\mathbf{Log}_{x^\lambda, x^{\alpha\lambda}}) \in \beta(\mathcal{C})\big).$$

The density of a basin is the probability that a logarithmically random starting value far above x first enters $[1, x]$ inside that basin. In the proofs of [17, 18], the law of Pass_x is controlled through the distribution of the Syracuse offsets modulo powers of 3. Problem 2(c) thus becomes a question about this 3-adic law on the small members of each basin.

Theorem 1 does not reach natural counting. For $3x + 1$, Mazur [19] has compared the first passage of the uniform law with that of the logarithmic law on the same block. His preprint is not yet refereed, but its proof chain is formalised in Lean 4; we therefore state the consequence as an implication.

Corollary 3 (Natural densities for $3x + 1$)**.** *Assume the transport estimate [19, Theorem 3.1]. Then every set $A \subseteq \mathbb{N}$ that is completely invariant under $T_{3,1}$ has a natural density, and it equals $\delta(A)$. In particular, the positive integers whose $3x+1$ orbit reaches 1 form a set with a natural density.*

Proof. Let $\alpha = 1001/1000$ as in [19]. For large N put $x = N^{1/\alpha^2}$ and $y = x^\alpha$; the odd integers in $[y, y^\alpha] = [N^{1/\alpha}, N]$ form a block of [19, Theorem 3.1]. Let $\mathbf{U}_y$ be uniform and $\mathbf{L}_y$ logarithmic on this block. By (14), by the scheduled no-hit bound and the ℓ^1 transport bound of [19, Theorem 3.1], and by the no-hit bound [17, (1.18)] for $\mathbf{L}_y$, which [19] uses with this α,

$$\mathbb{P}(\mathbf{U}_y \in A_1) = \mathbb{P}(\mathbf{L}_y \in A_1) + o(1) \qquad (N \to \infty).$$

Steps 4 and 6 give $\mathbb{P}(\mathbf{L}_y \in A_1) \to \delta(A)$: the window $[y, y^\alpha)$ splits into K consecutive windows of ratio $\alpha^{1/K} < 1 + \varepsilon_1$, and $\mathbb{P}(\mathbf{L}_y \in A_1)$ is their μ-weighted average; the endpoint conventions differ by at most one integer. The odd integers below $N^{1/\alpha}$ form a proportion $O(N^{1/\alpha - 1})$ of the odd integers below N. Hence A_1 has natural density $\delta(A)$ relative to the odd integers. Since $2^k m \in A$ if and only if $m \in A$, each set $\{2^k m : m \text{ odd}\}$ meets A in the same relative density, and A has natural density $\delta(A)$. □

Remark 6 (Natural densities for general c). Mazur's argument combines Tao's first-passage architecture with a discrepancy bound for the rotation by $\log_2 3$, which he derives from Rhin's irrationality measure [19]. For $T_{3,c}$ the rotation is the same, and the Syracuse offsets are c times those of $3x + 1$: in the notation of [17], $\mathrm{Syr}^n(m) = 3^n 2^{-|\vec{a}|} m + c\, F_n(\vec{a})$. An analogue of [19, Theorem 3.1] for $T_{3,c}$ would give natural densities for all basins of all maps $3x + c$ with $c \geq -1$, by the proof of Corollary 3, and would settle Problem 2(a). We have not verified that his argument carries over.

Theorem 1 settles existence in the logarithmic sense. Whether the natural densities exist for $c \neq 1$, and nontrivial bounds for their values, remain open; the stability up to 10^{10} is strong numerical evidence but no proof.

Problem 2. *By Proposition 5, the basin densities of $3x + c$ are determined by those of the cycles coprime to c' in the maps $3x + c'$, $c' \mid c$. For these cycles:*

- *(a)* Natural density. *Do the limits $\lim_{N\to\infty} d_N(\mathcal{C})$ exist? If so, they equal $\delta(\mathcal{C})$ by Theorem 1. For $c = 1$ they exist under the hypothesis of Corollary 3; Remark 6 outlines a route for general c.*
- *(b)* Explanation. *Which features of a cycle determine them: its parity word, the arithmetic position of its members, the distribution of its backward iterates?*
- *(c)* Prediction. *Is there an analytic or probabilistic model that predicts the observed values rather than fitting them?*

By Corollary 1, any answer to (b) and (c) must use global information. For $3x + 1$, the conjecture asserts a single basin of density 1.

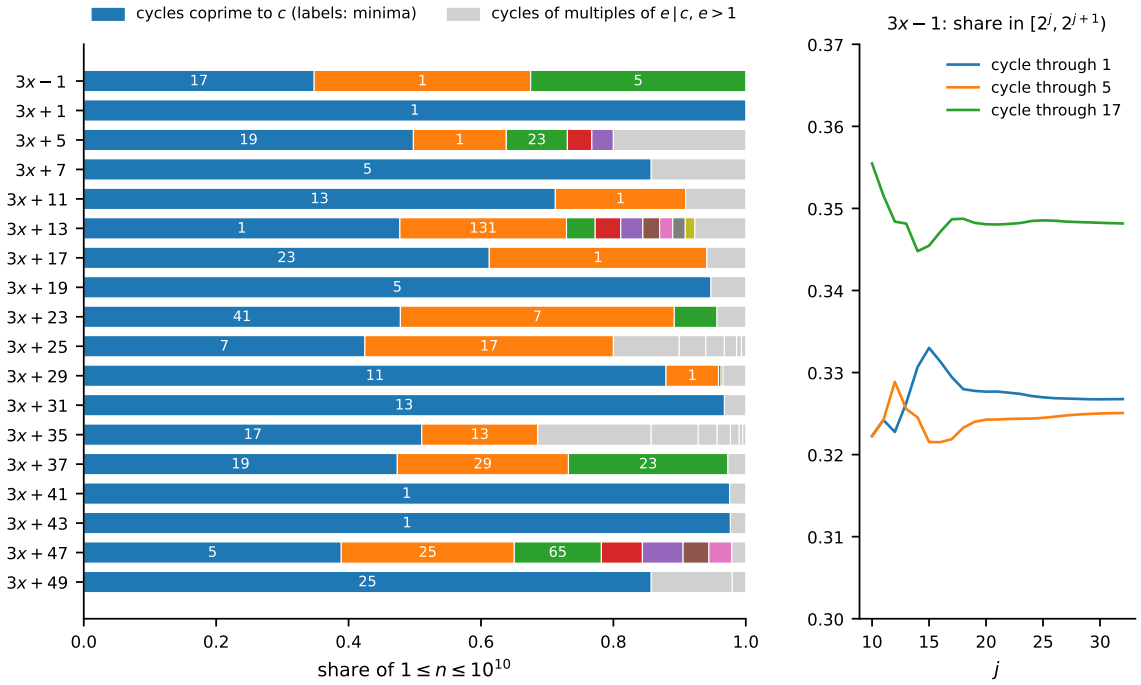


FIGURE 2. How the forest splits: empirical basin densities for $1 \le n \le 10^{10}$ (left) and the three basins of $3x-1$ per dyadic window (right).

6. Outlook

The local and the global view answer different questions. Cayley trees and their congruence classes describe how every forest of the family branches. Parity words and the numbers $2^l - q^k$ decide how many trees a forest has and where their cycles sit. We see five directions, in order of priority:

(1) *A theory of basin densities* (Problem 2): natural densities, most directly through an analogue of Mazur's natural-counting transport for $T_{3,c}$ (Remark 6); the features of a cycle that determine the values; and a predictive model, for instance through the law of first passage (Remark 5). This is the direction with the largest potential for new theory, independently of the Collatz conjecture itself.

(2) *The arithmetic structure of the primitive cycles* (Problem 1): why the equidistribution heuristic works and why the counts are underdispersed.

(3) *Longer words*: extending Proposition 4 to $l \le 44$ and beyond, and the role of the convergents of $\log_2 3$, such as $65/41$.

(4) *A Steiner-type theorem for Bařina's* $7x \pm 1$ [29]: no non-trivial circuits, via Diophantine approximation of $\log_2 7$ and Baker-type bounds, in the spirit of [3, 4].
(5) *Certification*: a Lean 4 formalisation of Lemma 1 and Propositions 1, 2 and 5 that refers to the maps themselves.

Code and data availability. The programs `census.c`, `census_all.c` and `basin.c`, the SLURM templates, the list of all 1097 primitive cycles as a CSV file, the basin counts with their dyadic windows, and the scripts for both figures are available in the folder *07 Basins of Attraction* of the repository [28].

Acknowledgments. We thank Dr. Kai Himstedt, Stephanie von Riegen and the whole EDIH Hamburg team for access to the Research Computing Infrastructure of the EDIH Hamburg.

Appendix A. Notation

Term	Definition
$T_{q,c}$	accelerated map: $(qx+c)/2$ for odd x, $x/2$ for even x
Orbit	the sequence $x, T(x), T^2(x), \ldots$
Cycle	a periodic orbit of distinct positive integers
Parity word B	parities of l consecutive iterates, 1 for odd
Type (l, k)	word length l (number of halvings) and number k of odd steps
D	$2^l - q^k$
$z_q(B)$	rotating-binary function (7); $x_B = c\, z_q(B)/D$
Necklace	rotation class of words; one necklace corresponds to one cycle
Branching number	even n with $(n - c)/3$ odd and not divisible by 3 (for $q = 3$)
Forest, tree, basin	all branching numbers; the predecessors of one cycle; the integers attracted by one cycle
Logarithmic density	$\delta(A) = \lim_{N\to\infty} \frac{1}{\log N} \sum_{n<N,\, n\in A} \frac{1}{n}$
First passage	$\mathrm{Pass}_x(m)$: the first Syracuse iterate of m that is at most x

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